universe has what amounts to a smallest possible size. This has profound consequences for our understanding of the universe at the moment of the bang itself, when the standard theory claims that its size has shrunk all the way to zero. Second, string theory has a small-radius/large-radius duality (intimately related to its having a smallest possible size), which also has deep cosmological significance, as we will see in a moment. Finally, string theory has more than four spacetime dimensions, and from a cosmological standpoint, we must address the evolution of them all. Let's discuss these points in greater detail.

In the Beginning There Was a Planck-Sized Nugget

In the late 1980s, Robert Brandenberger and Cumrun Vafa made the first important strides toward understanding how the application of these string theoretic features modifies the conclusions of the standard cosmological framework. They came to two important realizations. First, as we run the clock backward in time toward the beginning, the temperature continues to rise until the size of the universe is about the Planck length in all directions. But then, the temperature hits a maximum and begins to decrease. The intuitive reason behind this is not hard to come by. Imagine for simplicity (as Brandenberger and Vafa did) that all of the space dimensions of the universe are circular. As we run the clock backward and the radius of each of these circles shrinks, the temperature of the universe increases. But as each of the radii collapses toward and then through the Planck length, we know that, within string theory, this is physically identical to the radii shrinking to the Planck length and then bouncing back toward increasing size. Since temperature goes down as the universe expands, we would expect that the futile attempt to squeeze the universe to sub-Planck size means that the temperature stops rising, hits a maximum, and then begins to decrease. Through detailed calculations, Brandenberger and Vafa explicitly verified that indeed this is the case.

This led Brandenberger and Vafa to the following cosmological picture. In the beginning, all of the spatial dimensions of string theory are tightly curled up to their smallest possible extent, which is roughly the Planck length. The temperature and energy are high, but not infinite, since string theory has avoided the conundrums of an infinitely compressed zero-size starting point. At this beginning